

GODWIN TECH SOLUTION

Digital Solutions & Software Engineering

CAHIER DE CHARGES

(Functional & Technical Requirements Document)

SMART MARKET

Multi-Vendor E-Commerce Marketplace (Next.js / PWA)

Owned & Developed by

GODWIN TECH SOLUTION

Version 1.0

Date: 21/09/2026

Table of Contents

1. Context & Objectives

1.1 Context

Smart Market is a multi-vendor e-commerce marketplace developed by GODWIN TECH SOLUTION, built with Next.js and Progressive Web App (PWA) technology. The platform allows independent sellers to open online stores, publish products through paid subscription-based advertising plans, and complete secure transactions with buyers.

Customer payments are held in an escrow account, managed by the platform administrator, until both the buyer and the seller confirm that a transaction has been completed successfully.

This document defines the functional and technical scope of Smart Market as an internal product initiative led by GODWIN TECH SOLUTION, with no external client involved.

1.2 Objectives

- Create a scalable marketplace for multiple independent sellers.
- Allow merchants to create and manage their own online stores.
- Generate recurring revenue through subscription-based advertising packages.
- Secure every transaction through escrow payment management.
- Provide SEO-optimized pages to maximize organic search visibility.
- Integrate an AI chatbot to assist users and improve the overall customer experience.

1.3 Parties

Owner / Developer	GODWIN TECH SOLUTION
Project	Smart Market — Multi-Vendor E-Commerce Marketplace
Nature	Internal product initiative (no external client)
Platform	Web application built with Next.js, delivered as a Progressive Web App (PWA)
Document Version	1.0

2. User Roles & Permissions

Smart Market defines three primary user roles, each with a distinct set of capabilities on the platform.

2.1 Visitor

- Browse products and stores.
- Search products.
- Register an account.

2.2 Buyer

- Manage personal profile.
- Purchase products.

2.3 Seller / Store Owner

- Create a store.
- Purchase subscription plans.
- Publish advertisements.
- Manage products and sales.

3. Functional Scope — Platform Modules

The platform is organized into the following functional modules. Each module below describes what it does and how the different actors of the marketplace (visitors, buyers, sellers and administrators) interact with it.

Module 1 — Store Management

Each seller can create a public store containing:

- Store name.
- Logo and banner.
- Description.
- Location.
- Product categories.
- Contact information.

Every store must have SEO-friendly public pages.

Module 2 — Subscription & Advertisement Plans

Publishing advertisements on Smart Market requires an active subscription. Each subscription plan defines a duration, a number of advertisements allowed, and a set of available features.

Example plans:

- Starter: 30 days / 10 ads.
- Business: 90 days / 50 ads.
- Premium: 365 days / 200 ads.

When a subscription expires:

- Existing advertisements remain saved.
- Advertisements lose public visibility.
- The seller can renew existing ads, or publish new advertisements after renewing.

Module 3 — Advertisement Management

Each advertisement stores the following information:

- Title.
- Description.
- Images.
- Video.
- Price.
- Category.
- Location.
- Product condition.

Advertisement lifecycle:

Draft → Published → Active → Expired → Renewed → Archived

Module 4 — Search & SEO

SEO is a core requirement of the platform. Smart Market must provide:

- SEO-friendly URLs.
- Dynamic metadata.
- Product structured data.
- Store structured data.
- Sitemap generation.
- Optimized images.
- Category landing pages.
- Internal linking strategy.

Example URL patterns:

- smartmarket.com/product/product-name
- smartmarket.com/store/store-name

Module 5 — Marketplace Transaction Flow

1. The buyer selects a product.
2. The buyer completes payment.
3. Payment is transferred to the Smart Market escrow account.
4. The seller delivers the product or service.
5. The buyer confirms completion.
6. Funds are released to the seller.

Module 6 — Escrow Payment System

Smart Market acts as a trusted third party between buyer and seller.

Payment flow:

Buyer → Payment → Smart Market Escrow → Completion validation → Seller payout

The system must support:

- Payment tracking.
- Commission calculation.
- Seller balance management.
- Transaction history.

Module 7 — AI Chatbot**Customer assistance:**

- Answer common questions.
- Guide users through the purchasing process.
- Explain subscription plans.
- Assist sellers.

Marketplace assistance:

- Product discovery.
- Recommendation assistance.
- Order information support.

Administrative assistance:

- Basic reporting support.
- Customer service assistance.

Module 8 — Administration

Super Admin:

- Full platform control.
- Escrow management.
- Subscription management.
- User management.
- Platform configuration.

Accountant:

- Financial reports.
- Payment monitoring.
- Seller payouts.
- Transaction reconciliation.

Customer Service:

- User assistance.
- Complaint management.
- Dispute handling.

Module 9 — Seller Dashboard

Dashboard indicators:

- Active advertisements.
- Expired advertisements.
- Subscription status.
- Views.
- Orders.
- Sales revenue.

Module 10 — Admin Dashboard

Key indicators:

- Users.
- Stores.
- Revenue.
- Subscription sales.
- Escrow balance.
- Transactions.
- Disputes.

Module 11 — Notifications

Supported channels:

- Email.

- Push notifications.
- In-platform notifications.

Triggering events:

- Subscription expiration.
- Advertisement expiration.
- New orders.
- Payment confirmation.
- Dispute creation.

4. Non-Functional Requirements

- Performance: pages should load within 2–3 seconds under normal network conditions.
- Availability: target uptime of 99% for the platform and backoffice, dependent on the hosting infrastructure.
- Security: encrypted connections (HTTPS/SSL), role-based access control, secure password storage, and careful handling of payment-related data.
- Usability: intuitive, mobile-first navigation with a minimal number of steps to publish a listing or complete a purchase.
- Scalability: the architecture must support growth in sellers, listings and concurrent transactions without requiring a redesign.
- SEO: pages must be indexable and optimized for organic discovery from launch.

5. Technical Architecture

Layer	Technology
Frontend	Next.js with Progressive Web App (PWA) support
Backend	Scalable REST API architecture
Database	Relational database management system (RDBMS)
Authentication	Secure authentication with role-based permissions
Payments	Payment gateway integration with escrow management
Storage	Cloud storage for product images, videos and store assets
AI Chatbot	Conversational AI service integrated across the marketplace experience

6. Data Model — Main Entities

Entity	Purpose
Users	Visitors, buyers, sellers and administrators registered on the platform.
Stores	Public storefronts created and managed by sellers.
Subscription Plans	Catalog of paid plans defining duration, ad quota and features.
Subscriptions	Active or past subscription purchased by a seller.
Advertisements	Product listings published by sellers, following the ad lifecycle.
Categories	Classification structure used for products, stores and SEO landing pages.
Orders	Purchases made by buyers against advertisements.
Payments	Payment records associated with orders and subscriptions.
Escrow Transactions	Funds held by the platform between payment and delivery confirmation.
Wallets	Seller balances resulting from completed transactions and payouts.
Reviews	Feedback left by buyers on products, stores or sellers.
Disputes	Complaints or conflicts raised by buyers or sellers, handled by Customer Service.
Notifications	Email, push and in-platform alerts triggered by platform events.
Roles	Named permission sets (Super Admin, Accountant, Customer Service, etc.).
Permissions	Granular actions granted to a given role.

7. Development Roadmap

The roadmap below groups the platform's functional scope into three delivery phases. It reflects functional priority only — detailed timelines, resourcing and budget will be defined separately once the roadmap is confirmed.

Phase	Scope
Phase 1 — Core Platform	Authentication, Stores, Subscription system, Advertisement management, Marketplace, Escrow payment.
Phase 2 — Engagement & Insights	AI chatbot, Reviews, Advanced analytics, Promotions.
Phase 3 — Intelligence & Automation	AI recommendations, Advanced marketplace automation.

8. Acceptance Criteria

The platform is considered functionally complete when:

- Users can create accounts.
- Sellers can create stores.
- Sellers can subscribe to plans.
- Advertisement quotas are enforced.
- Expired advertisements become invisible to the public.
- Buyers can purchase products.
- Escrow payments work correctly end-to-end.
- Funds are released to sellers after confirmation.
- SEO requirements are implemented.
- The AI chatbot is operational.

9. Document Validation

This document defines the agreed functional and technical scope of the Smart Market platform as an internal initiative of GODWIN TECH SOLUTION. Any change to this scope will be reflected in a revised, re-validated version of this Cahier de Charges.

Role	Name	Signature & Date
Prepared by	Product & Engineering — GODWIN TECH SOLUTION	
Reviewed by		
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